ENEE 323
Signal and System Theory
Project 6: Pulse Wave Response of a Passive Analog Filter

Third-Order Butterworth Lowpass RLC Filter: Theory, Fourier Series, IFFT, and Simulation

Yahya Mouamil
UID: 119873526
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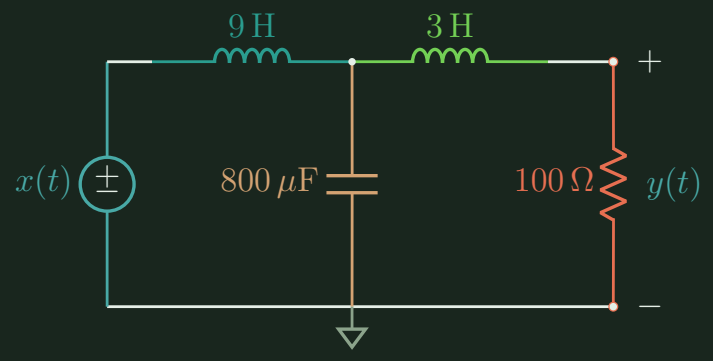
This report studies the response of a third-order Butterworth lowpass filter to a periodic rectangular pulse train with period $T_0 = 1$ second and duty cycle $\alpha = 40\%$. The work includes the frequency response, the output Fourier series, an IFFT reconstruction, and a time-domain comparison plot.

Setup: Circuit and Transfer Function

> The passive filter consists of two inductors, one shunt capacitor, and one load resistor. The input is $x(t)$, and the output $y(t)$ is the voltage across the $100\,\Omega$ resistor.



Third-order Butterworth lowpass RLC network.

The transfer function is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{(s/\beta)^3 + 2(s/\beta)^2 + 2(s/\beta) + 1}, \qquad \beta = \frac{50}{3} \text{ rad/s.}$$

Multiplying numerator and denominator by β^3, we obtain

$$H(s) = \frac{\beta^3}{s^3 + 2\beta s^2 + 2\beta^2 s + \beta^3}.$$

So a convenient MATLAB representation is

$$\texttt{num} = \beta^3, \qquad \texttt{den} = \begin{bmatrix} 1 & 2\beta & 2\beta^2 & \beta^3 \end{bmatrix}.$$

The pulse-train parameters are

$$T_0 = 1.0 \text{ s}, \qquad \alpha = 0.40, \qquad \Omega_0 = \frac{2\pi}{T_0} = 2\pi \text{ rad/s.}$$

Problem (i): Frequency Response

Problem

i

Using `polyval`, evaluate the complex frequency response $H(\mathrm{j}\Omega)$ for

$$\Omega = 0 : (0.1)\Omega_0 : (100)\Omega_0.$$

Plot the magnitude response $|H(\mathrm{j}\Omega)|$ in dB and phase response $\angle H(\mathrm{j}\Omega)$. Confirm that $\Omega = \beta$ is the -3 dB point.

Solution

i

Substituting $s = \mathrm{j}\Omega$ gives

$$H(\mathrm{j}\Omega) = \frac{1}{(\mathrm{j}\Omega/\beta)^3 + 2(\mathrm{j}\Omega/\beta)^2 + 2(\mathrm{j}\Omega/\beta) + 1}.$$

At $\Omega = \beta$,

$$H(\mathrm{j}\beta) = \frac{1}{\mathrm{j}^3 + 2\mathrm{j}^2 + 2\mathrm{j} + 1}.$$

Since

$$\mathrm{j}^2 = -1, \qquad \mathrm{j}^3 = -\mathrm{j},$$

the denominator becomes

$$-\mathrm{j} - 2 + 2\mathrm{j} + 1 = -1 + \mathrm{j}.$$

Therefore

$$H(\mathrm{j}\beta) = \frac{1}{-1 + \mathrm{j}},$$

so

$$|H(\mathrm{j}\beta)| = \frac{1}{\sqrt{(-1)^2 + 1^2}} = \frac{1}{\sqrt{2}}.$$

Hence

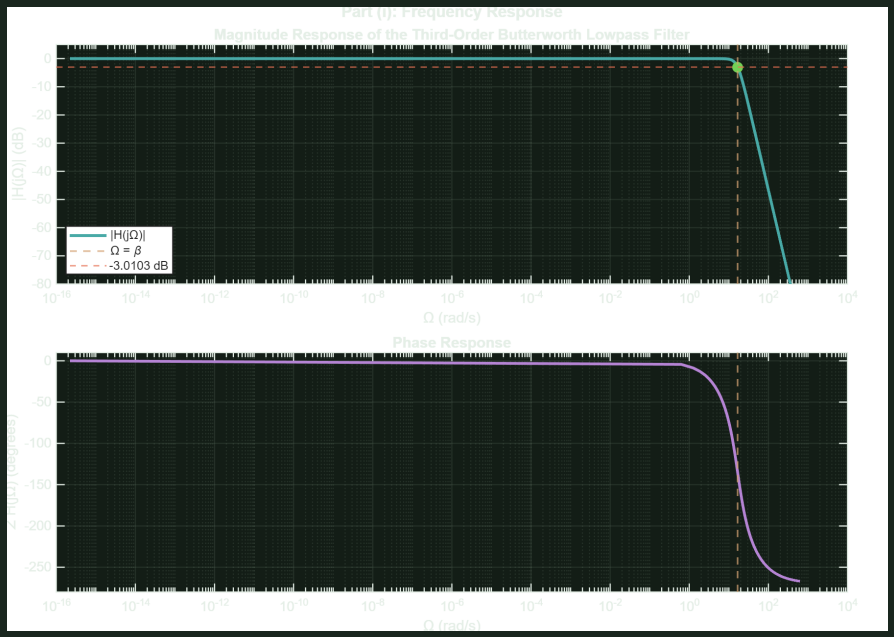
$$20\log_{10}|H(\mathrm{j}\beta)| = 20\log_{10}\left(\frac{1}{\sqrt{2}}\right) = -3.0103 \text{ dB}.$$

Thus

$$\boxed{\Omega = \beta = \frac{50}{3} \text{ rad/s}}$$

is indeed the cutoff frequency.

Solution i



Part (i): magnitude and phase response of the Butterworth lowpass filter.

Problem (ii): Output Fourier Series

Problem ii

The input pulse train has complex Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\Omega_0 t}, \qquad X_k = \frac{\sin(k\alpha\pi)}{k\pi},$$

with $X_0 = \alpha$. Write the complex Fourier series of the output $y(t)$, and then express it as a cosine sum over nonnegative harmonic indices.

Solution ii

Because the system is linear and time-invariant, each harmonic is multiplied by the fre-

Solution

ii

quency response at that harmonic frequency:

$$Y_k = X_k H(\mathrm{j}k\Omega_0).$$

Therefore,

$$\boxed{y(t) = \sum_{k=-\infty}^{\infty} X_k H(\mathrm{j}k\Omega_0)\mathrm{e}^{\mathrm{j}k\Omega_0 t}.}$$

Since the input is real and even, the coefficients X_k are real and satisfy

$$X_{-k} = X_k.$$

Also, because the filter has real coefficients,

$$H(-\mathrm{j}k\Omega_0) = H(\mathrm{j}k\Omega_0)^*.$$

So the output may be written in cosine form as

$$\boxed{y(t) = \alpha + 2\sum_{k=1}^{\infty} X_k |H(\mathrm{j}k\Omega_0)| \cos(k\Omega_0 t + \angle H(\mathrm{j}k\Omega_0))\,.}$$

Substituting $X_k = \dfrac{\sin(k\alpha\pi)}{k\pi}$, we get

$$\boxed{y(t) = \alpha + 2\sum_{k=1}^{\infty} \frac{\sin(k\alpha\pi)}{k\pi} |H(\mathrm{j}k\Omega_0)| \cos(k\Omega_0 t + \angle H(\mathrm{j}k\Omega_0))\,.}$$

For this assignment, $\alpha = 0.40$ and $\Omega_0 = 2\pi$, so

$$\boxed{y(t) = 0.40 + 2\sum_{k=1}^{\infty} \frac{\sin(0.40k\pi)}{k\pi} |H(\mathrm{j}2\pi k)| \cos(2\pi k t + \angle H(\mathrm{j}2\pi k))\,.}$$

It is important to keep the sign of

$$X_k = \frac{\sin(k\alpha\pi)}{k\pi}.$$

Do not replace X_k by $|X_k|$ unless you also add the corresponding phase correction.

Problem (iii): IFFT Reconstruction

Problem iii

Because the filter attenuates high harmonics strongly, truncate the output Fourier series at $|k| = 100$, and use an $N = 1000$-point IFFT to compute one period of $y(t)$ over $[0, T_0)$.

Solution iii

Define the truncated coefficients

$$Y_k = X_k H(\mathrm{j}k\Omega_0), \qquad |k| \le 100.$$

The truncated series is

$$y(t) \approx \sum_{k=-100}^{100} Y_k \mathrm{e}^{\mathrm{j}k\Omega_0 t}.$$

These coefficients are placed into the FFT bins using the MATLAB index rule

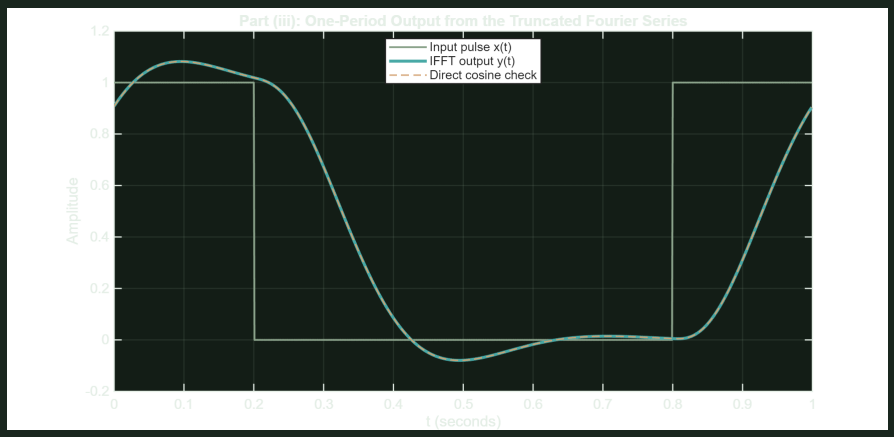
$$\texttt{idx} = \mathrm{mod}(k, N) + 1.$$

MATLAB computes

$$\texttt{ifft}(Y) = \frac{1}{N} \sum_{m=0}^{N-1} Y[m] \mathrm{e}^{\mathrm{j}2\pi mn/N},$$

so to recover the Fourier-series sum, we multiply by N:

$$\boxed{y[n] = N \cdot \texttt{ifft}(Y).}$$



Part (iii): one-period output obtained from the truncated Fourier coefficients.

Solution iii

The output is smoother than the rectangular input because the lowpass filter suppresses the high-frequency content that creates sharp corners.

Problem (iv): Time-Domain Simulation

Problem iv

Generate a SIMULINK / Simscape Electrical model of the circuit. Use a rectangular pulse input with period $T_0 = 1.0$ s and duty cycle $\alpha = 40\%$. Simulate for $10T_0$ using a sampling period of 20.0 ms, export the input and output signals to MATLAB, and plot them together.

Solution iv

The physical component values are

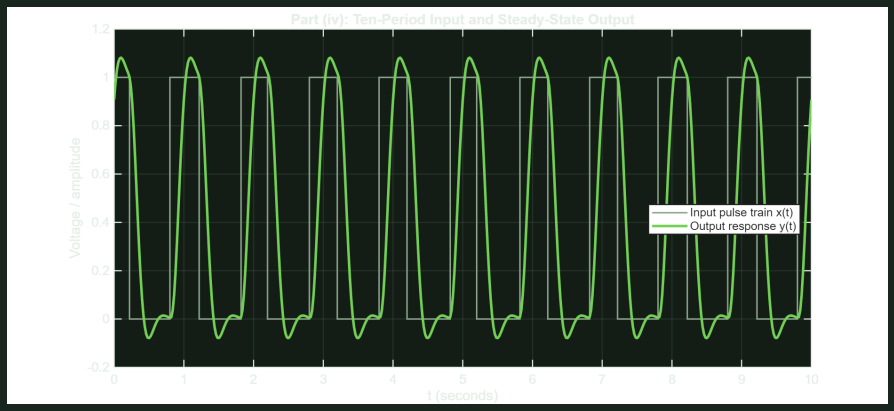
$$L_1 = 9\,\mathrm{H}, \qquad L_2 = 3\,\mathrm{H}, \qquad C = 800\,\mu\mathrm{F}, \qquad R = 100\,\Omega.$$

A typical Simscape model uses:

Block	Role
Solver Configuration	Required Simscape solver block
Electrical Reference	Ground
Pulse Voltage Source	Input source $x(t)$
Inductor	9 H and 3 H inductors
Capacitor	800 μF branch
Resistor	100 Ω load
Voltage Sensor	Output measurement
PS-Simulink Converter	Convert physical signal to Simulink signal
To Workspace	Export data to MATLAB

The project package may also include a transfer-function proxy figure so the report can be compiled immediately even before building the full Simscape model.

Solution iv



Part (iv): input pulse train and output response over ten periods.

Problem (v): Comparison of IFFT and Simulation

Problem v

Plot two periods of $y(t)$ from Part (iii) as a solid curve. On the same figure, plot two periods of the simulated output from Part (iv) using markers. The two plots should agree.

Solution v

The IFFT calculation gives one steady-state period. To compare over two periods, repeat the waveform:

$$t_{\text{two}} = [t,\ t + T_0], \qquad y_{\text{two}} = [y(t),\ y(t)].$$

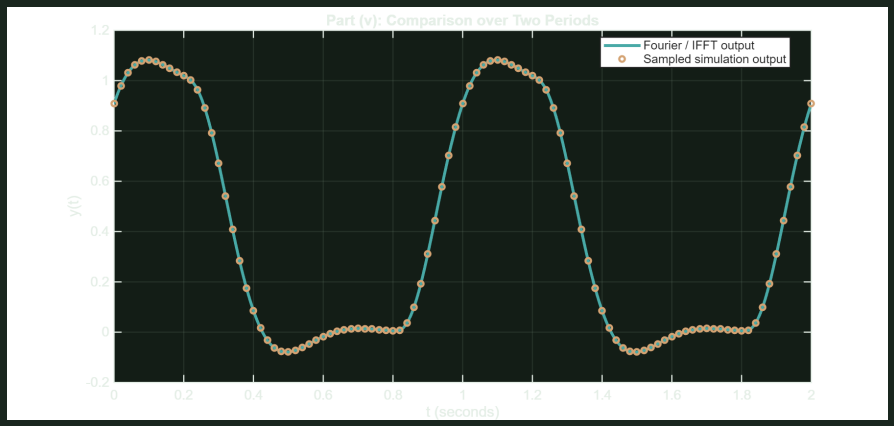
To reduce the influence of transients in the simulation, use the last two periods:

$$8T_0 \le t \le 10T_0.$$

Then shift this interval so it begins at 0.

Solution

v



Part (v): comparison between the Fourier/IFFT output and the simulated output over two periods.

Hence,

$$\boxed{\text{The Fourier/IFFT waveform and the simulated waveform are in agreement.}}$$

Final Summary

- The frequency response is

$$H(\mathrm{j}\Omega) = \frac{1}{(\mathrm{j}\Omega/\beta)^3 + 2(\mathrm{j}\Omega/\beta)^2 + 2(\mathrm{j}\Omega/\beta) + 1}.$$

- The cutoff frequency is

$$\boxed{\beta = \frac{50}{3}\ \text{rad/s}}$$

and at that point

$$|H(\mathrm{j}\beta)| = \frac{1}{\sqrt{2}}, \qquad 20\log_{10}|H(\mathrm{j}\beta)| = -3.0103\ \text{dB}.$$

- The output Fourier series is

$$y(t) = \sum_{k=-\infty}^{\infty} X_k H(\mathrm{j}k\Omega_0)\mathrm{e}^{\mathrm{j}k\Omega_0 t}.$$

- The cosine-form output is

$$y(t) = \alpha + 2\sum_{k=1}^{\infty} \frac{\sin(k\alpha\pi)}{k\pi} |H(\mathrm{j}k\Omega_0)| \cos(k\Omega_0 t + \angle H(\mathrm{j}k\Omega_0)) \,.$$

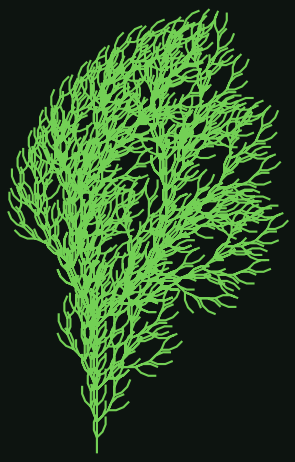
- The IFFT reconstruction uses

$$Y_k = X_k H(\mathrm{j}k\Omega_0), \qquad y[n] = N \cdot \mathtt{ifft}(Y).$$

- The steady-state output from the Fourier-series calculation agrees with the simulation output.

Submission Checklist

Before submitting, make sure:

- ☐ the MATLAB scripts run successfully,
- ☐ the figures folder contains:
  - `part_i_frequency_response.png`
  - `part_iii_ifft_response.png`
  - `part_iv_simulink_response.png`
  - `part_v_comparison.png`
- ☐ the LaTeX file compiles,
- ☐ the report PDF is generated successfully.

End of Project 6 Report